SIMROAA

Multi-Agent Recommendation System for Recommending Accessible Learning Objects

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***Abstract*— This full paper presents the Multi-Agent Recommendation System for Recommending Accessible Learning Objects developed to attend students with disabilities and professors' computation area. The system was proposed based on the qualitative analysis of the questionnaires applied and answered by the professors of the area, to classify, organize and provide the most appropriate Learning Objects according to professors' preferences. In this way, the professor informs the type of impairment (visual, physical, hearing and/or cognitive), discipline and content, based on these requirements, receives the best recommendations of the intelligent agent. The methodology that we use is Multiagent Systems Engineering. The web page was developed using the Laravel Framework PHP, the data repository was created using the MySQL database, and to perform the implementation we used the Java Agent Development Framework. As a result of the research, we present a recommendation system of accessible learning objects using intelligent agents and communities of practice through the recommendation mechanism based on trust. The results obtained through the implementation of the system provide a new motivational practice trough accessible educational content, which contributes significantly to the teaching-learning process of the computation course and to inclusive education.**

Keywords- Students with Disabilities; Accessible Learning Objects; Multi-Agent System.

I. INTRODUCTION

Currently, bringing technology into the classroom is one of the biggest challenges faced by educators. In this sense, education goes through a continuous process of transformation, several approaches are discussed to create an innovative, inclusive and authentic environment, that supports the infusion of new technologies in learning.

Nowadays, there are several discussions in evidence to make learning more meaningful, enjoyable, effective and inclusive. Training Computer Science professors is a crucial task in addressing the inequity in computer science education[1].

Experts who design and develop Learning Objects (LO) have suggested: "that adopting a System for Recommending may provide an efficient vehicle for reuse of Learning Objects stored in a data repository" [2],[3],[4] and [5]. Thus provisioning of tools to recommender learning objects, like the one presented in this article, can increase interest, improve adoption, encourage the content creation and subsequent reuse.

The model presented one contribution to the Inclusive Model proposed by [2], with a case study [3] and it is related within of open educational resource initiatives which include authoring tools and repositories of learning objects [4].

Teaching environments based on multi-agent architectures allow supporting the development of systems more powerful, fast and of low cost [5]. In this context, the agents promoted the recovery and recommendation of Accessible Learning Objects (ALO) to professors, in accordance with the requirements.

Our paper is structured as follows. In this section, the objectives of this research are introduced, followed by a review of the motivations for the research (Section 2). Section 3 presents the system. Section 4 presents the results and discussions, and the section closes with an overview (Section 5).

II. STATE OF ART

A. Learning Object (LO)

The IEEE defines Learning Objects as "any entity, digital or non-digital, which can be used, re-used or referenced during technology supported learning". As examples of Learning Objects include: multimedia and instructional content, learning objectives, instructional software and software tools, persons, organizations, or events referenced during technology supported learning [6].

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Learning objects serve as an instructional design component in e-learning for the development and delivery of educational content [7].

For Collis [8] the technology involving learning objects can be considered at four levels. In this research, we considered the technology related to the repository in which the objects are collected, including database technology and/or learning content management technology [9].

Furthermore, is important to highlight that repositories and reuse of learning objects in different learning environments have a description of standardized content, called metadata.

The IEEE organizations [9] and Instructional Management System (IMS) [10] consider the following characteristics as common to learning objects: reusability, portability, modularity, metadata, and interactivity.

A learning object must have its granularity proper in order to provide reuse, must be developed for use on various platforms (through this, respecting the feature of portability), and be produced in independent and not sequential modules, and finally being used in conjunction with other resources.

The most common metadata standards are Learning Object Metadata (LOM) of the IEEE and the Sharable Content Object Reference Model (SCORM).

B. Accessible Learning Object (ALO)

According to the IMS Global Learning Consortium [10] accessibility is defined as a set of efforts in order to facilitate access to the media and social, educational and cultural characteristics to reduce the effect of a limitation of the environment and greater equality for students with disabilities. For the World Wide Web Consortium (W3C) [11], accessibility is an extension of sites, products, services or information, making them available to the greatest number and variety of people regardless of their physical and motor, cultural and social limitations and perceptions. In this sense, both have worked to put guidelines and standards in place to assist in ensuring interoperability and accessibility.

Considering the above concepts, it is understood that an Accessible Learning Object is a Learning Object with accessibility requirements and social inclusion premises of People with Disabilities (PWD). For this reason, it is possible to provide autonomy in access to information and learning through digital resources. In this research, the concept of Accessible Learning Objects was adopted according to [2] as any digital material (images, videos, web pages, animations or simulations), since they provide information for the construction of knowledge, attends standards and guidelines of accessibility, and was structured so that they can be reused.

Batanero [12] explores in your paper the relationship between accessible learning objects and accessible web content. Macedo [13] presents a significant contribution to teams or developers working on the development of Accessible Learning Object.

C. Multi-agent Systems (MAS)

There are several multi-agent-based e-learning applications in the literature [14]. In this way, the authors state that artifacts represent resources and tools and can be dynamically created and managed by agents to support their respective activities.

According to Russel [15], an agent is all that can be considered capable of perceiving its environment through sensors and acting on this environment through actuators. For Wooldridge [16] an agent is a computer system, situated in some environment, which is capable of performing autonomous action in this environment in order to meet the objectives for which it was designed.

A Multi-Agent System is a distributed system where a group of autonomous agents reaches their goals in a reactive, proactive and social way. According to [17], an MAS is a system composed of cooperative or competitive agents, which interact with each other to achieve a common goal or individual.

The agent is a computer system positioned in an environment that can act with autonomy flexible to reach the objectives for which it was designed [18]. For the author, some properties must be observed in an agent. In accordance with [19], we highlight features as situatedness, autonomy, pro-activeness, adaptiveness, responsiveness, mobility.

As described in [20], there are several types of agents, and they can be: software or hardware, stationary or mobile, persistent or temporary, reactive, or cognitive. In this sense, the design of a multi-agent system should consider several aspects as type or problem, the autonomy of the agent, knowledge representation, communication protocols, definition of system organization, data modeling, cooperation, and conflict resolution.

III. RECOMMENDATION SYSTEM

Mizquero [22] define Recommendation Systems (RS) as a tool that aims to provide users with useful information according to their requirements, delivering those items closer than expected. The recommendation systems aim to return items from queries, predicting a possible interest of the user seeking to avoid overloading and exhaustive search for information. Recommendation models include content-based, collaborative, hybrid strategies [23].

It's an important highlight that intelligent education systems have been the subject of research application in the different teaching modalities.

In this paper, we proposed a recommendation environment architecture that supports education by utilizing intelligent agents. This model explores the application of artificial intelligence and smart learning environments. Like this, we intend to meet pedagogical demand through technological tools to support the process of teaching and learning.

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The model focuses in to accomplish the recommendation of accessible learning objects to professors according to the requirements: discipline, content, and disability. SIMROAA is not a public repository, it is undergoing updating and testing to meet a recent demand from the Computing Nucleus of the State University of Amazonas when it consolidates we intend to make it public. Currently, there are ten OAA that was developed by students and professors through case studies, reported and described by Mourão and Netto [2018a, 2018b, 2019]. The goal is that in the future we can have an expressive number of objects so that their metadata can be registered and shared on the Internet.

SIMROAA is the recommendation system that was developed to recommend the most appropriate objects to the professor, where intelligent agents accomplish this communication process. Fig. 1 shows the storage repository named as ROAA (Repository of Accessible Learning Objects) and all communication between the intelligent agents.

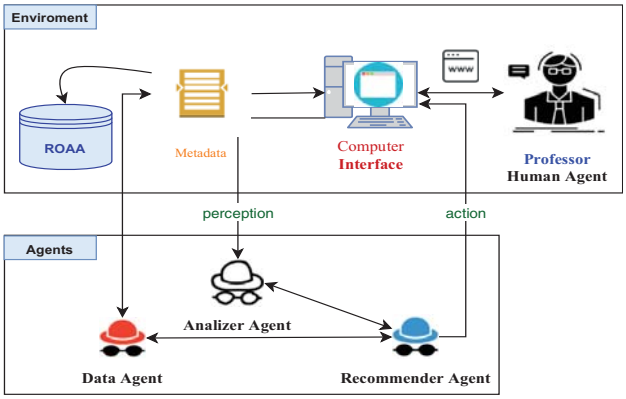


Fig. 1. Scheme of communication between agents of the SIMROAA

As can be seen in Fig. 1, professors should initially log in Web page produced for this purpose, with their institutional email to gain access to the recommendation of the objects. This scheme we have the ROAA Repository that is used for ALO storage, which has been developed according to the SCORM (Sharable Content Object Reference Model) standard [2], [3]. The work of [30] presents an accessible learning object, following the model of [3] applied to the teaching of mathematics. Through the information provided in the repository metadata, the agent can compare the information of the ALO with the requirements.

For our purposes agents interact with each other to meet an overall goal. Norvig and Russel [21] define the agent function as the mapping of a sequence of perceptions to an action. In this sense, each agent has his responsibilities of perception and action in this context.

Therefore, we have one human agent and three types of technological agents: recommender, analyzer, and data, which were as designed shown in Fig. 1, and defined below.

A. Human Agent

This Agent entitled "Human-Agent" has as role send your requirements by a web page, and aims receive the best recommendation of learning objects. The human agent is a computer professor. Fig. 2 shows the SIMROAA Sign in screen, which provides access to the Recommendation System.

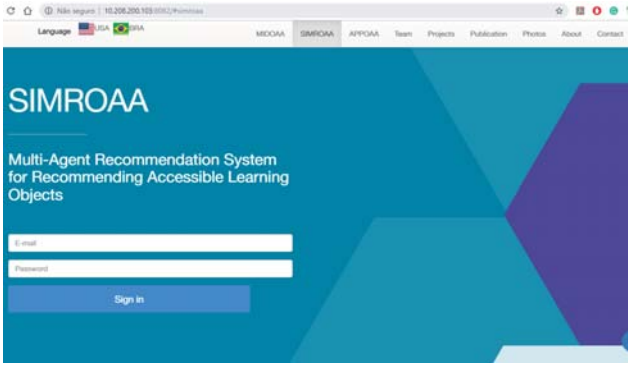


Fig. 2. SIMROAA - OAA Sign In Screen.

Fig. 3 shows the interface where the professor reports the requirements needed to obtain the objects. The professor must inform the type of disability (auditory, visual, cognitive or multiple) and has the option to obtain an object without defining the type of disability. If there is a disability, the professor should inform the level of Disability (for example Deafness Moderate, Severe, Severe, Deep or Anacusis).



Fig. 3. SIMROAA - OAA Request Screen.

B. Analyzer Agent

This Agent entitled "Analyzer Agent" has a role to perceive the environment, and uses all of the history and knowledge to perform its function. Therefore, the agent only to accomplish an action from a perception.

First, the agent receives and analyzes the data provided by the professor in relation to the course, discipline, content, and disability (physical, auditory, cognitive, visual or multiple) of the student. After, the agent recommends ALO the most appropriate resource. Followed, the agent sends a message with the requirements for the recommendation agent.

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C. Recommender Agent

This Agent entitled “Agent Recommender” has a role to recommend for professors the Accessible Learning Objects. After receiving the message of the analyzer agent, the recommendation agent confirms receipt. Then, it communicates with the data agent to obtain a response in relation to the request. The data agent runs the search in the repository by means of the metadata and sends the results to the Recommendation Agent.

Recommendation Agent will provide to the Human Agent (professor) a list containing one (1) or more ALO in accordance with its request or, if not any ALO be found, it will issue a warning message (Fig. 4). The items are a list by type of disability and content when found.

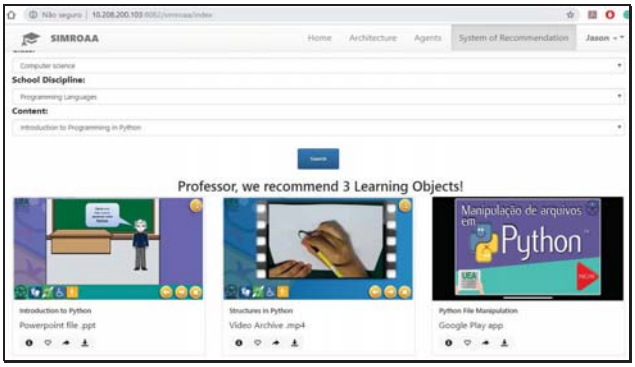


Fig. 4. SIMROAA - OAA Recommendation of the Learning Objects. Pictures (In Portuguese). Example of ALO Recommendation – The First picture is the Introduction to Python, the second is Structures in Python, and the third is Python File Manipulation.

D. Data Agent

This Agent entitled “Data Agent” is responsible for recover accessible learning objects in repository ROAA. The access occurs through the metadata, data are collected, and sent to Recommender Agent. The agent organizes, classifies the items, and sends the results obtained for the human agent. If there are no results a message is sent stating "that no objects were found in accordance with the requested requirements".

According to this approach, an agent is capable to play the role of people and can acquire new knowledge by the interaction with professors and other agents, raising the potential their understanding and performance. The ALO metadata permits the identification of what educational topic is related to the requirements.

In this way, the agents can find out what is the content or disability associated with the user experience shown by the professor, and then to show complementary information or recommend other objects.

SIMROAA makes possible the recommendation of Accessible Learning Objects automatically through the environment Web.

Fig. 5 show the general architecture system and the key layers. The first layer is a presentation layer that shows the Web Interface which provides the communication between the human agent and the web page. In this sense, professor and system interact, and after the professor send the solicitation, the intelligent agents if communicate with each other in the agents' layer. Furthermore, the data and infrastructure layer is the ROAA data repository and its metadata. Metadata stores the records of learning objects, making it more efficient for objects search.

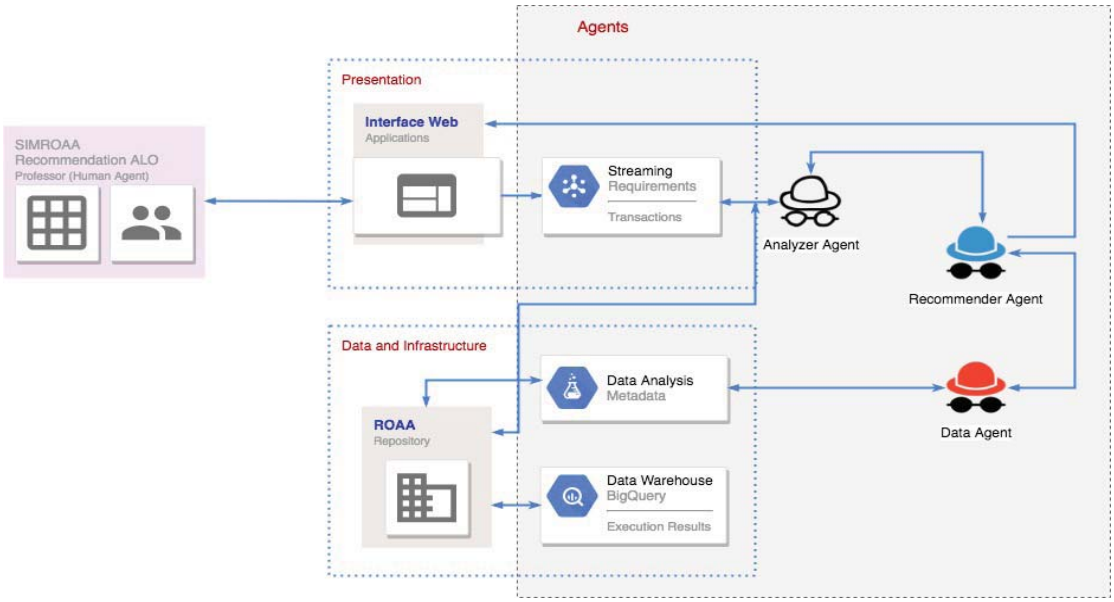


Fig. 5. SIMROAA Architecture.

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To evaluate the proposed approach were used accessible learning objects developed by students and professors to the computation course proposed by [2],[3]. Finally, we presented the results of the evaluation, discussed current limitations, and future work directions.

IV. Results and Discussions

SIMROAA has the intention to submit a new proposal aimed at professors of undergraduate courses in computing, which often reuse materials or educational resources, but do not find in one place, learning objects developed in accordance with standards and accessibility guidelines.

The objective of this proposal is to meet a recent demand for students with disabilities, providing professors with the use and recommendation of an appropriate educational resource. This project was designed after the application of a questionnaire with the professors to know the importance and availability of the use and reuse of educational resources aimed at attending students with disabilities.

Of this computing core, out of a total of 23 professors, 15 answered. Therefore, 87% expressed that they agreed with the possibility of having a repository of educational resources, produced by the professors or specialists of the area and that they could be shared among the group of professors. Fig. 6 expresses the result.

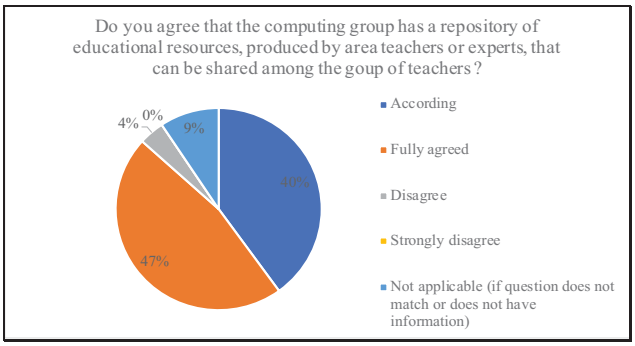


Fig. 6. Questionary: Question 1 about Repository.

The questionnaire was applied to verifying the viability of the development of the repository and multi-agent system. When asked, if they would like to have access for an intelligent system that recommended learning objects in accordance with requirements the disability students, 44% expressed full agreement and 56% agreed. Fig. 7 expresses the search result.

We believe that the result was expressive and shows not only the viability but also the interest and willingness of professors to contribute to the institution's inclusive education.

Thus, after the viability analysis, the project was started using the concept of multi-agents.

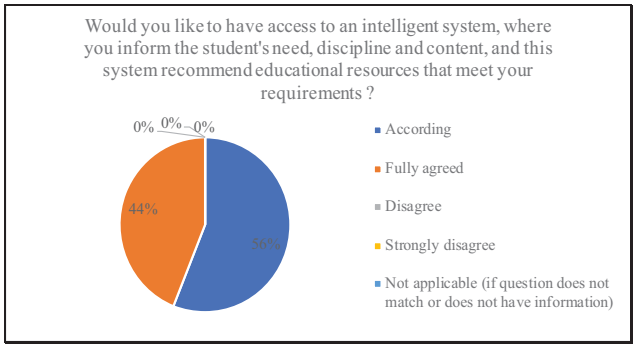


Fig. 7. Questionary: Question 2 about Multi-agent System.

For this, we defined that the recommendation system would adopt the content-based model that verifies the similarity of new objects with objects that a user liked previously. However, it is necessary to obtain the characteristics of the object and to compare them with the user preference [24]. For modeling, was used the Multiagent Systems Engineering (MASE) methodology. The MASE is a focused approach to the paradigm of agents and assists the stages of analysis, design, and development of multi-agent systems based on the principles of software engineering [25].

The web application has been developed using the Laravel Framework PHP. Also, the forms used Google Forms tools, whose license was acquired by the institution. The data repository was created using the MySQL database that has a good performance, reliability, and ease of use, whose administration is through the phpMyAdmin application. To perform the implementation, execution, and testing of the system multi-agent was used the framework Java Agent Development Framework (JADE), and the Java programming language, then was made the connection to the database and communication tests with the agents.

Tests of the recommendation system were divided into 3 stages:

1. *Send:* The professor Sign In the Web Site, with his institutional email, which is validated by the system to guarantee its authenticity;

2. *Communication between Agents*: Agents perceive the sending of the request, and then the analyzer agent exports the data from the Database and communicates the recommendation agent, which sends the information to the data agent. This analyzes and returns the result, then the recommender agent show for the professor.

3. *Reception*: the message or object is received by the professor.

After the communication tests between the agents, it was verified the viability of recommending the accessible learning objects, using the tools and methodologies mentioned above. Based on the agents presented in Fig. 8, it was possible to design and implement the system agents. Fig.8 shows the exchange of messages between agents, using the JADE Sniffer tool [29].

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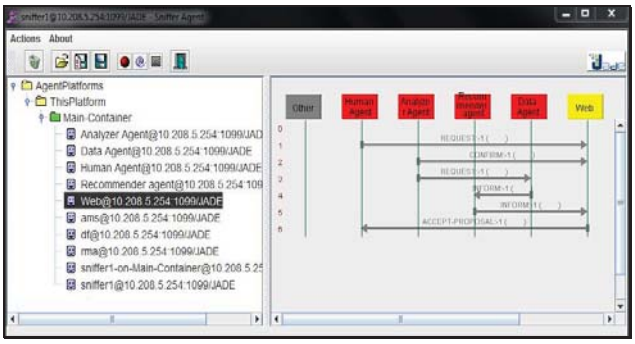


Fig. 8. Visualization of the agents exchanging message, through the Sniffer tool.

The collaboration between agents, inference and learning capabilities are interesting features [36] when we realized the tests. It also occurred the effective communication and exchange of messages between agents simulated in the environment. However, we perceive that some particular requirements of trust in the recommendations of learning objects did not meet expectations. In the analysis performance and confiability requirements reached 80%, and availability ranged from 75 to 90%.

Considering the presented scenario, we searched for recommendation models that could help increase the confiability of the object recommendation to make our system more robust. Thus, we adopt the model cited by Molina, Koutrika, and Parameswaran [26]. The authors affirm that the Communities of Practice (COP) are experiencing a flood of data with the increasing volume of digital objects. The article reports a recommendation mechanism using COP's in the format of learning objects, in communities of virtual practices, through a network or circle of trust.

Training professors to teach computer science through sustained support by a community of practice [31], [32], [33], is a challenge to be overcome in computing due to the technical training of professors [1]. Equally important, reflect in the affirmation of your professional self-identity in the classroom [34].

Thus, we accomplish prototyping and simulation to improve the trust indexes necessary to guarantee a precise recommendation, considering the architecture shown in Fig.5. In SIMROAA, the agent of recommendation is responsible for filtering the relationships of trust and classifying the relevant learning objects according to a human agent, in our case the professor. In this way, the model was contextualized in the Communities of Practice in the field of teaching and learning, where trust relationships are essential for the development of collaboration between professors. The proposed model used the trust construct among professors as a metric for the recommendation of learning objects.

According to [27] COP is a combination of three elements: the domain of knowledge, which defines a set of issues; the group of people who care about this domain; and shared practice for the internalization of knowledge. That is, if there is trust among its members, then the communities produce faster and more prosperous results. Table 1 shows the trust-based recommendation model.

The interactions at the COP are shown in Table 1 by crossing the components of the model such as the Human Agent (professors), Recommendation Agent, Learning Object and the interactive environment that is the COP.

TABLE 1 INTERACTIONS OF THE TRUST BASED ON THE RECOMMENDATION MODEL.

	Components			Environment
	Human Agent (Professor)	***Accessible Learning Object***	***Recommendation Agent***	***Virtual practical community COP***
Human Agent (Professor)	Indicates the degree of confidence	Develop, use and reuse objects.	Receives recommendation of the Analyzer Agent, and analyze professor profile;	Sends requirements and assigns trust value, participating and collaborating with the environment.
Accessible Learning Object	-	It is associated with a disability, and pedagogical content;	It is recommended by a technology agent.	Repository of Accessible Learning Objects (ROAA)
Recommendation Agent	-	-	Communicate and collaborate	Acts in the environment making the recommendation

It is important to emphasize that establishing trust among the members (professors) of a COP is an accepted social process as a prerequisite for cooperation. For [28] levels of trust among members influence the sharing of resources and the exchange of useful knowledge.

In this framework, the recommendation agents interact by presenting accessible learning objects relevant to human agents. Suggestions from the recommendation agents are related to the various decision-making processes.

Professors indicate the confidence index of the learning objects. This indication is marked in the user interface, where a Likert scale of 7 points represents the confidence degree. Graphically it is shown the scale as 7 stars above the professor avatar. This confidence index is used to indicate the reliability degree between people and their confidence circles.

The inclusion of the index (icon favorite) is justified by the fact that professors are more likely to use the accessible learning objects produced by their trust network. Thus, the recommendation mechanism performs clustering of the trusted network, removing nodes with indexes below the scale means from the graph. After the procedure, the trusted items were identified, and the agent then sorts the ALO, classifying them according to the user context information.

Recommendation agent clusters the network to detect the hidden structures and group the trusted members. In this context, people are represented by vertices and the trust are the weighted edges that connect the members of the network.

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Each edge of the network has a weight relative to a weighted graph G = {V, E}, where the confidence index of the vertex V'1 with the vertex V'2 expresses the weight. The confidence network is formed by the way made to the weighted graph where the vertices are removed, whose edges have a weight less than 0.4 or unreliable within the Likert scale of 1 to 7. Table 2 shows the scheme.

TABLE 2 TRUSTED MATRIX IN THE FUNCTION OF THE SPECIFIC PROFESSOR.

Human Agent	Professor 1	Professor 2	Professor 3	Professor 4
Professor 1	0.2	0.6	0.7	X
Professor 2	0.4		X	0
Professor 3	0.3	X	0.6	0.3
Professor 4	X		0.3	0.5

Thus, we obtain a sub graph G '= {V', E '} representing the relations dependable as a function of a specific human agent. The direct connections are important, in this example represents the confidence in only 1 degree in the graph G '.

In the graph G', the data agent queries accessible learning objects produced by members of the trusted network, and route the information to the recommendation agent that to accomplish the classification and recommendation.

The information used refers to the user login (professor), registered by the analyzer agent. The recommendation agent, after consulting the data agent in the repository, presents the resulting accessible learning objects according to the professor profile and informed requirements.

ALO resulting from the query are classified according to the weight assigned to it (considering the author-professor). In this way, the degrees of confidence of the weighted edges adjacent to the professor who produced the ALO are summed. Considering the process, a weight is assigned to each learning object stored in the repository. The weight is used as the ordering criterion when presented to the human agent.

In this way, we obtained a functional recommendation system, using intelligent agents and COP, where the recommendation is based on trust. It's important to create an instrument to identify factors related to professors and classes, in order to plan actions for training and definitions of teaching strategies [37].

System of Recommendation was evaluated according to the criteria of usability and accessibility, through a form made available to the professors that use the system, applying Likert scale metrics. The form aims to improve the system's functionalities, with the aim of achieving a degree of accuracy, confiability, and availability. Therefore, the recommendation system considers the accuracy of the system the result obtained by the evaluation of the user in relation to the content that was recommended to him and the note of the prediction of the recommendation system. In this sense, the system expressed 93% accuracy.

It was assumed that the objects to be recommended to the users are only those that have prediction equal or superior to 4 points, within a Likert scale of 7 points, in agreement with the established confidence level. In this sense, Figure 9, presents the result of the usability evaluation accomplishes.

Usability Evaluation corresponds to the quality of Recommendation System with regard to veracity, precision and balanced presentation of ideas. The items used in the evaluation are based on the Nielsen Heuristics [38]. Fig. 9 below shows the results obtained in this evaluation.

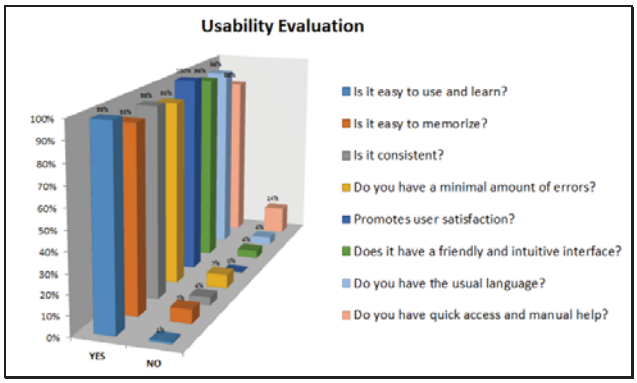


Fig. 9. Usability Evaluation of the SIMROAA.

In relation to the Usability Evaluation analysis, 100% of interviewed affirmed that the system promotes user satisfaction. Also, 96% related that the interface of the system is simple and intuitive; have a usual language, and are consistent. Another 93% affirmed that there is tolerance of errors and have a minimal amount of errors. The quick access and manual help presented one icon that promotes the contact. In relation to the General Evaluation analysis the Fig. 10 below shows the results obtained about the system.

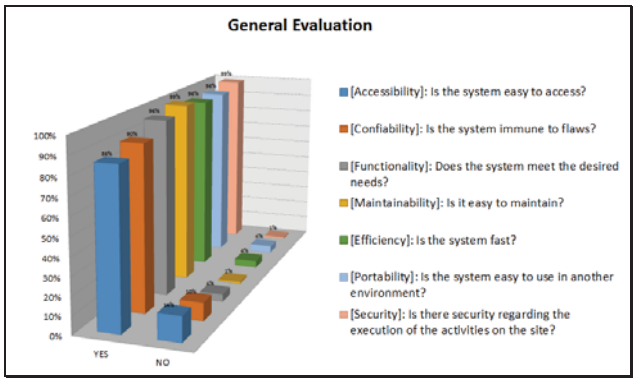


Fig. 10. General Evaluation of the SIMROAA.

In order to evaluate the adherence of the recommendation system to the levels of acceptable requirements, the standard ISO 9126 was followed for product quality that involves

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important non-functional requirements such as confiability, functionality, maintainability, efficiency, portability, and safety.

Website of the System of recommendation to General Evaluation shows that there about 96% of interviewed affirmed that the functionality, efficiency, and portability system have the same concepts. Thus, in relation to the functionality requirement, the System does what is appropriate, interacts with systems properly, complies with laws and standards, and prevents unauthorized access to data. As for efficiency, the system has an excellent response time and execution speed; allows for easy modification, is easy to modify, adapt and test. In the item portability, it was noticed its fail adaptation, installation in other environments and its capacity of substitution. Also, 90% related that the system responds to failures and presents data recovery. Another 99% affirmed that the system is easy for maintenance.

In relation to the Accessibility Evaluation analysis, 86% of the evaluators report that the images help in understanding the content and that they have explanatory text. However, the system shows a description for non-text content, and use colors and fonts appropriately.

V. CONCLUSION

This paper presents the development of the SIMROAA that is a recommendation system designed to attend the Model proposed by Mourão e Netto [2]. The objective of this proposal is to provide the professors of the computing nucleus of the State University of Amazonas an educational solution to the recommendation of accessible educational resources. And, in the future, allow the repository to be available and can be shared by other universities, professors, and courses.

Models and support mechanisms were analyzed to make this recommendation process. The viability of the methodology was obtained through tests with agents, prototypes, and simulations. Besides, most professors considered motivating to learn through learning objects, as well as considered that the objects presented a new reality.

Recommendation systems provide access to the elements and provide distinct strategies for recommending objects. In this educational context, its use prevents professors from spending a lot of time locating accessible learning objects to attend their goals. The use of COP allowed validating the trust, which is a fundamental element for the construction of relations between the members of a community. The inclusion of the trust construct to the metrics guarantees the degree of trust of the neighbors identified in the professor's network of relationships. Thus, the mechanism also depends on the social system to indicate the degree of trust and the technological agents accomplish clustering tasks of confidence circles and the association for inference of accessible learning objects.

The results obtained by this work contribute to support cooperation and collaboration in teaching and learning processes. This way, encourage the professors to use educational tools that collaborate with improvements in the teaching and learning process, stimulating the adoption of inclusive education.

As future work, we propose that this recommendation system after consolidated will become public and can be used by other higher education institutions and professors. Also, explore and incorporate data mining techniques within ROAA. And finally, the development of a chatterbot, where the users will interact with the application.

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